$C_{qe}^{\text{i1\_i2\_i3\_i4\_}} \to$

$$\frac{1}{16\pi^2}\Big(-\frac{2}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{d}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{108}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{d}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{d}}^{p}]$$

$$\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{27}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{e}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{36}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{e}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{8}{135}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{e}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{l}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{5}{72}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{l}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{135}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{l}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{216}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{u}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{5}{27}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{u}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{32}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{u}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{27}\,g_1^4\,\mathrm{LF}_{3,0}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{18}\,g_1^4\,\mathrm{LF}_{4,-1}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{8}{135}\,g_1^4\,\mathrm{LF}_{5,-2}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{36}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_1,\,m_{\tilde{e}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,\,m_{\tilde{e}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{18}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,\,m_{\tilde{e}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,\,m_{\tilde{e}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{36}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_1,\,m_{\tilde{e}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,\,m_{\tilde{e}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{18}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,\,m_{\tilde{e}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,\,m_{\tilde{e}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_1,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{648}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_1,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{648}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}[m_2,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,2,-1}[m_2,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{24}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}[m_2,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-2}[m_2,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}[m_2,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,2,-1}[m_2,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{24}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}[m_2,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-2}[m_2,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_3,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}[m_3,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_3,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}[m_3,\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_3,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}[m_3,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_3,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}[m_3,\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{9}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{p},\,\tilde{\mu}]\,\delta_{i3i4}+\frac{1}{18}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{2,2,-1}[m_{\tilde{d}}^{p},\,\tilde{\mu}]\,\delta_{i3i4}+$$

$$\frac{1}{18}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{p},\,\tilde{\mu}]\,\delta_{i3i4}+\frac{1}{18}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde{e}}^{i3},\,m_1]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{36}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde{e}}^{i3},\,m_1]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{18}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde{e}}^{i4},\,m_1]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{36}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde{e}}^{i4},\,m_1]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{18}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{2,1,0}[m_{\tilde{l}}^{p},\,\tilde{\mu}]\,\delta_{i1i2}-$$

$$\frac{1}{36}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{2,2,-1}[m_{\tilde{l}}^{p},\,\tilde{\mu}]\,\delta_{i1i2}-\frac{1}{36}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{l}}^{p},\,\tilde{\mu}]\,\delta_{i1i2}+$$

$$\frac{1}{648}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1},\,m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1},\,m_1]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{24}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1},\,m_2]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1},\,m_2]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1},\,m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1},\,m_3]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{648}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2},\,m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{1296}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2},\,m_1]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{24}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2},\,m_2]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2},\,m_2]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2},\,m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2},\,m_3]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{9}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{2,1,0}[m_{\tilde{u}}^{p},\,\tilde{\mu}]\,\delta_{i3i4}-\frac{1}{9}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{2,2,-1}[m_{\tilde{u}}^{p},\,\tilde{\mu}]\,\delta_{i3i4}-$$

$$\frac{1}{9}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{u}}^{p},\,\tilde{\mu}]\,\delta_{i3i4}+\frac{1}{18}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{2,1,0}[\tilde{\mu},\,m_{\tilde{d}}^{p}]\,\delta_{i3i4}+$$

$$\frac{5}{36}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},\,m_{\tilde{d}}^{p}]\,\delta_{i3i4}-\frac{1}{36}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},\,m_{\tilde{d}}^{p}]\,\delta_{i3i4}-$$

$$\frac{1}{36}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{2,1,0}[\tilde{\mu},\,m_{\tilde{l}}^{p}]\,\delta_{i1i2}+\frac{5}{36}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},\,m_{\tilde{l}}^{p}]\,\delta_{i1i2}-$$

$$\frac{1}{18}\,g_1^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},\,m_{\tilde{l}}^{p}]\,\delta_{i1i2}-\frac{1}{9}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{2,1,0}[\tilde{\mu},\,m_{\tilde{u}}^{p}]\,\delta_{i3i4}-$$

$$\frac{1}{36}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},\,m_{\tilde{u}}^{p}]\,\delta_{i3i4}-\frac{1}{36}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},\,m_{\tilde{u}}^{p}]\,\delta_{i3i4}-$$

$$\frac{1}{72}\,g_1^4\,\mathrm{LF}_{2,1,1,-1}[m_1,\,m_{\tilde{e}}^{i3},\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}[m_1,\,m_{\tilde{e}}^{i3},\,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{72}\,g_1^4\,\mathrm{LF}_{2,1,1,-1}[m_1,\,m_{\tilde{e}}^{i4},\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}\,g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}[m_1,\,m_{\tilde{e}}^{i4},\,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{4}\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\overline{y_e}^{ri3}\,y_e^{ri4}\,\mathrm{LF}_{2,1,1,-1}[\tilde{\mu},\,m_{\tilde{d}}^{p},\,m_{\tilde{l}}^{r}]-$$

$$\frac{1}{2}\,\tilde{\mu}^2\,\overline{y_e}^{pi3}\,y_e^{pi4}\,\overline{y_u}^{i2r}\,y_u^{i1r}\,\mathrm{LF}_{2,1,1,0}[\tilde{\mu},\,m_{\tilde{l}}^{p},\,m_{\tilde{u}}^{r}]\Big)$$